The Sun loses material from its surface at a rate of more than a million tons per second. We call this the "solar wind," which takes the form of high-energy charged particles. Traveling up to a thousand miles per second, these particles stream through space and are deflected by planetary magnetic fields. The particles spiral down toward the north and south magnetic poles, forcing collisions with gas molecules and leaving the atmosphere aglow with colorful aurora. The Hubble Space Telescope has spotted aurora near the poles of both Saturn and Jupiter. And on Earth, the aurora borealis and australis (the northern and southern lights) serve as intermittent reminders of how nice it is to have a protective atmosphere.

Earth's atmosphere is commonly described as extending dozens of miles above Earth's surface. Satellites in "low" Earth orbit typically travel between one hundred and four hundred miles up, completing an orbit in about ninety minutes. While you can't breathe at those altitudes, some atmospheric molecules remain—enough to slowly drain orbital energy from unsuspecting satellites. To combat this drag, satellites in low orbit require intermittent boosts, lest they fall back to Earth and burn up in the atmosphere. An alternative way to define the edge of our atmosphere is to ask where its density of gas molecules equals the density of gas molecules in interplanetary space. Under that definition, Earth's atmosphere extends thousands of miles.

Orbiting high above this level, twenty-three thousand miles up (one-tenth of the distance to the Moon) are the communications satellites. At this special altitude, Earth's atmosphere is not only irrelevant, but the speed of the satellite is low enough for it to require a full day to complete one revolution around Earth. With an orbit precisely matching the rotation rate of Earth, these satellites appear to hover, which make them ideal for relaying signals from one part of Earth's surface to another.

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Newton's laws specifically state that, while the gravity of a planet gets weaker and weaker the farther from it you travel, there is no distance where the force of gravity reaches zero. The planet Jupiter, with its mighty gravitational field, bats out of harm's way many comets that would otherwise wreak havoc on the inner solar system. Jupiter acts as a gravitational shield for Earth, a burly big brother, allowing long (hundred-million-year) stretches of relative peace and quiet on Earth. Without Jupiter's protection, complex life would have a hard time